MILITARY SPECIFICATION

MECHANISM, SINGLE POP-UP TARGET SCORING

This specification has been approved by the Office of Naval Research (Naval Training Device Center), Department of the Navy.

1. SCOPE

1.1 This specification covers one type of electro-mechanical device to position a target and provide a readout of hits for small arms firing exercises.

1.1.1 This item has been designated by the Office of Naval Research (Naval Training Device Center), as Device 3C52E for Navy procurement (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-corner

Military

MIL-P-116	Preservation; Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Training Device Center

DL 662500	List of Drawings of Single Pop-Up Target Scoring Mechanism, Device 3C52E

FSC 6920

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be as specified on drawings of 2.1.

3.2 Design.- The mechanism as shown on the drawings of 2.1 shall provide:

(a) A holder for a small arms target which can be moved from a concealed position to a firing position and returned to the concealed position

(b) A counter which will register the hits made on the target.

3.2.1 Interchangeability.- All parts, subassemblies and assemblies having the same part number shall be directly and completely interchangeable with each other with respect to installation and performance, regardless of sources from which obtained, without alteration and without selection for fit or performance.

3.3 Construction.- Construction shall be as shown on the drawings of 2.1.

3.3.1 Watertightness.- The Control Box Assembly (Part No. 662501) and the Mechanism Box Assembly (Part No. 662510), shown on drawings of 2.1, shall be so constructed that no water shall enter these assemblies when they are submerged in water to a depth of one foot of water over the highest point of the assembly enclosures. The cord entry connecters on the control box assembly shall be provided with removable closure seals to prevent water entry.

3.4 Performance.-

3.4.1 Power input.- The mechanism shall be operable from single-phase alternating current of 115 volts, plus or minus 10 percent, at a frequency of 60 hertz, plus or minus 5 percent. Current consumption shall not exceed 2 amperes.

3.4.2 Operation.-

3.4.2.1 Manual operation.- The mechanism, with Government-furnished Target C8426438 installed, shall be capable of moving the target to the horizontal or vertical position when controlled by the operator. The rate of motion shall be equal to 20 cycles per minute (cpm) plus or minus 2 cpm (a cycle consists of movement of the target from the vertical position to the horizontal position and back to the vertical position). The mechanism shall register hits upon impact of a force of 20 inch/ounces, without lowering the target when manually controlled to the vertical position with the motor switch placed in the off position.

3.4.2.2 Automatic operation.- The mechanism, with Target C8426438 installed, shall register hits and move the target from the vertical position to the horizontal position and back to the vertical position when the target is hit by the force of impact specified in 3.4.2.1. The rate of motion shall be the same as that specified in 3.4.2.1.

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3.4.2.3 Hit counting.- The mechanism shall be provided with a counter having four digit wheels which can be manually reset to zero. The counter shall be capable of registering hits at a rate of 600 per minute. The counter shall register the total number of hits, plus or minus 2 percent.

3.4.3 Endurance.- The mechanism, with Target C8426438 installed, shall operate continuously at a temperature of 75 degrees Fahrenheit, plus or minus 10 degrees, for a minimum of 4800 cycles without any depreciation in its performance.

3.5 Details of components.- The mechanism shall consist of the following major components, the details of which shall be as shown on the drawings of 2.1:

(a) Control Box assembly

(b) Mechanism box assembly

(c) Hit switch cable assembly

(d) Target holder assembly

3.6 Dimensions and tolerances.- Dimensions and tolerances shall be as shown on the drawings of 2.1. Tolerances, not shown on the drawings of 2.1, shall be such as to assure meeting the interchangeability and performance requirements specified in 3.2.1 and 3.4.

3.7 Finish and color.- Finish and color shall be as shown on the drawings of 2.1.

3.8 Nameplates or product markings.-

3.8.1 Nameplate.- A Government-furnished nameplate shall be affixed to the mechanism box assembly in the location shown on the drawings of 2.1.

3.9 Government-furnished property.- The Government will furnish 2 each of Target C8426438, and one U. S. Naval Training Device Center nameplate.

3.10 Workmanship.- The mechanism shall be constructed and finished with particular attention given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, sanding, grinding, painting, riveting, machine-screw assemblage, welding and brazing.

3.10.1 Cleanup.- Units shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material after final assembly.

3.10.2 Wire stripping.- Wire stripping shall be done without nicking or otherwise damaging the wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial

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laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual, electrical, and mechanical examinations and testing of materials, subassemblies, parts, and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements specified in this specification.

4.4 Quality conformance inspection.-

4.4.1 Preproduction inspection.- Preproduction inspection shall be conducted on the preproduction model and shall consist of all the examinations and tests specified in 4.5 and 4.6.

4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-accepted preproduction model.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Dimensions and tolerances

(d) Finish and color

(e) Nameplates or product markings

(f) Workmanship

(g) Preparation for delivery.

4.6 Tests.- The following tests shall be performed:

(a) Watertightness (see 3.3.1)

(b) Input power (see 3.4.1)

(c) Manual operation (see 3.4.2.1)

(d) Automatic operation (see 3.4.2.2)

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(e) Hit counting (see 3.4.2.3)

(f) Endurance (see 3.4.3)

4.7 Test methods.-

4.7.1 Watertightness test.-

4.7.1.1 Initial test.- Prior to conducting the tests specified in 4.7.3, 4.7.4 and 4.7.5 both the control box assembly and the mechanism box assembly shall be submerged in water having a temperature of 75 degrees Fahrenheit, plus or minus 10 degrees, to demonstrate conformance to 3.3.1, for a period of one hour.

4.7.1.2 Final test.- After completion of the tests specified in 4.7.3, 4.7.4, and 4.7.5, the test specified in 4.7.1.1 shall be repeated to demonstrate conformance to 3.3.1.

4.7.2 Input power test.- The input power voltage shall be set at the low tolerance limit, at the operating voltage and at the high tolerance limit specified. 3.4.1 and a record shall be made of the current used while performing the tests specified in 4.7.3, 4.7.4, and 4.7.5 to demonstrate conformance to 3.4.1.

4.7.3 Manual operation test.- The target shall be cycled by manual operation of the switches 10 times at each voltage specified in 4.7.2. The average time required for transition of the target from the horizontal to the vertical position and from the vertical to the horizontal position shall be recorded. The target shall be hit by a one-ounce weight, swung through a 90 degrees, plus or minus 10 degrees, angle on a radius of 20 inches to demonstrate conformance to 3.4.2.1.

4.7.4 Automatic operation test.- The target control switch shall be placed in the up position, and the target shall be hit by the weight specified in 4.7.3, 10 times at each voltage specified in 4.7.2 and the average cycle time shall be recorded to demonstrate conformance to 3.4.2.2.

4.7.5 Hit counting test.- The target shall be hit by the weight specified in 4.7.3, 25 times at the low and high voltages and 50 times at the operating voltage specified in 4.7.2 and the number of counts registered shall be recorded to demonstrate conformance to 3.4.2.3.

4.7.6 Endurance test.- The device shall be cycled continously to demonstrate conformance to 3.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C (see 6.2.).

5.1.1 Level A.- For Level A, each mechanism shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

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5.1.2 Level C.- For Level C, preservation and packaging shall conform to manufacturer's commercial practice.

5.2 Packing.- Packing shall be level A or C (see 6.2.).

5.2.1 Level A.- Each unit packaged in accordance with 5.1.1 shall be packed in an exterior container conforming to PPP-B-621.

5.2.2 Level C.- Units packaged in accordance with 5.1.2 shall be packed in containers that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a matter designed to prevent damage due to shock, vibration or impact during shipment, handling and storage.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The Single Pop-Up Target Scoring Mechanism supporting a suitable target will be used for small arms firing practice to move the target from a concealed position to the firing position and register hits made.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Procuring activity designation (device number or other) (see 1.1.1).

(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Definitions.- Except as follows, definitions will be furnished upon written request to the procuring activity.

6.3.1 Preproduction model.- A production item furnished under production contracts for the purpose of determining compliance with requirements of the detail training device specification already established and accepted. It shall be electrically and mechanically representative of production sets though not necessarily made with production line jigs, fixtures and methods.

6.3.2 Production unit.- A unit in its final mechanical and electrical form of final production design made by production tools, jigs, fixtures and methods.

Custodian:
Navy-TD

Preparing activity:
Navy-TD

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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